

Optimization Techniques — Assignment Report

Continuous & Discrete Optimization using Metaheuristics

Team Details

Roll Number : CS23I1026
Contact : 7058077596
Course : Optimization Techniques
Submitted to : Dr. Venkatesh
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Chosen Problems & Methods

	Problem	Methods
Problem A (Continuous)	Optimal Design of Welded Beam Structure	Adaptive PSO, SA with Restart, Differential Evolution
Problem B (Discrete)	Optimal Bin Packing Problem	Genetic Algorithm, SA with Reheating, Tabu Search

1 Introduction

Optimization finds the best feasible solution with respect to an objective. Real-world engineering and combinatorial problems often have nonlinear, constrained, or discrete search spaces where classical gradient methods fail. Metaheuristics offer robust alternatives requiring no gradient information. This report solves two problems: **Problem A** (Welded Beam Design — continuous, 4 variables, 5 constraints) and **Problem B** (Bin Packing — discrete, NP-hard). Each is solved with **three metaheuristics** compared over **30 independent runs** with convergence plots, boxplots, and sensitivity analysis.

2 Problem A: Welded Beam Design (Continuous Optimization)

2.1 Mathematical Formulation

A steel beam welded to a rigid wall must be designed to minimize manufacturing cost while satisfying structural constraints. Decision variables $\mathbf{x} = [h, l, t, b]$ are weld thickness, weld length, beam thickness, and beam width.

Variable bounds:

Var.	Description	Unit	LB	UB
h	Weld thickness	in	0.1	2.0
l	Weld length	in	0.1	10.0
t	Beam thickness	in	0.1	10.0
b	Beam width	in	0.1	2.0

Objective function (minimize manufacturing cost):

$$f(\mathbf{x}) = 1.10471 h^2 l + 0.04811 t b (14 + l) \quad (1)$$

Inequality constraints ($g_i(\mathbf{x}) \leq 0$):

$$g_1 = \tau(\mathbf{x}) - 13600 \leq 0 \text{ (shear stress)}, \quad g_2 = \sigma(\mathbf{x}) - 30000 \leq 0 \text{ (bending stress)} \quad (2)$$

$$g_3 = h - b \leq 0 \text{ (geometric)}, \quad g_4 = \delta(\mathbf{x}) - 0.25 \leq 0 \text{ (deflection)} \quad (3)$$

$$g_5 = P - P_c(\mathbf{x}) \leq 0 \text{ (buckling)} \quad (4)$$

Supporting equations ($P = 6000 \text{ lb}$, $L = 14 \text{ in}$, $E = 30 \times 10^6 \text{ psi}$, $G = 12 \times 10^6 \text{ psi}$):

$$\tau = \sqrt{\tau'^2 + \frac{\tau' \tau'' l}{R} + \tau''^2}, \quad \tau' = \frac{P}{\sqrt{2} h l}, \quad \tau'' = \frac{M R}{J}, \quad M = P \left(L + \frac{l}{2} \right) \quad (5)$$

$$R = \sqrt{0.25(l^2 + (h + t)^2)}, \quad J = 2\sqrt{2} h l \left(\frac{l^2}{12} + \frac{(h+t)^2}{4} \right) \quad (6)$$

$$\sigma = \frac{6 P L}{b t^2}, \quad \delta = \frac{4 P L^3}{E b t^3}, \quad P_c = \frac{4.013 E \sqrt{t^2 b^6 / 36}}{L^2} \left(1 - \frac{t}{2L} \sqrt{\frac{E}{4G}} \right) \quad (7)$$

Penalty function for constraint handling:

$$F(\mathbf{x}) = f(\mathbf{x}) + 10^6 \cdot \sum_{i=1}^5 [\max(0, g_i(\mathbf{x}))]^2 \quad (8)$$

2.2 Algorithm Implementation

2.2.1 Adaptive PSO

PSO [3] maintains 40 particles. Each represents $\mathbf{x} = [h, l, t, b]$ and flies through search space using personal best $\mathbf{p}_i$ and global best $\mathbf{g}$. **Enhancement** — **Adaptive inertia**: w decreases linearly $0.9 \rightarrow 0.4$, promoting exploration early and exploitation late:

$$w(t) = 0.9 - 0.5 \cdot \frac{t}{T}, \quad \mathbf{v}_i \leftarrow w \mathbf{v}_i + c_1 r_1 (\mathbf{p}_i - \mathbf{x}_i) + c_2 r_2 (\mathbf{g} - \mathbf{x}_i), \quad \mathbf{x}_i \leftarrow \mathbf{x}_i + \mathbf{v}_i \quad (9)$$

Params: $n = 40$, $T = 500$, $c_1 = c_2 = 2.0$. *Why PSO?* Gradient-free, works on real-valued vectors, collective memory guides search.

2.2.2 Simulated Annealing with Restart

SA [4] accepts worse neighbors with probability $e^{-\Delta f/T}$, cooling as $T \leftarrow \max(\alpha T, T_{\min})$. **Enhancement** — **Restart**: After 500 stagnant iterations, restart from a new random point (warm $T \leftarrow T_{\text{init}}/2$), retaining global best. *Params*: $T_{\text{init}} = 1000$, $\alpha = 0.995$, $\text{max_iter} = 5000$.

2.2.3 Differential Evolution (DE/rand/1/bin)

DE [2] is state-of-the-art for continuous engineering problems. Per generation:

$$\text{Mutation: } \mathbf{v}_i = \mathbf{x}_{r_1} + F(\mathbf{x}_{r_2} - \mathbf{x}_{r_3}), \quad r_1, r_2, r_3 \text{ distinct, } \neq i \quad (10)$$

$$\text{Crossover: } u_{ij} = \begin{cases} v_{ij} & \text{rand}() < CR \text{ or } j = j_{\text{rand}} \\ x_{ij} & \text{otherwise} \end{cases} \quad (11)$$

$$\text{Selection: } \mathbf{x}_i \leftarrow \mathbf{u}_i \text{ if } F(\mathbf{u}_i) \leq F(\mathbf{x}_i), \text{ else keep } \mathbf{x}_i \quad (12)$$

Params: pop= 40, $F = 0.8$, $CR = 0.9$, max_iter= 500. *Why DE?* Self-adaptive step size from population differences gives superior convergence on constrained non-convex problems.

2.3 Results and Convergence

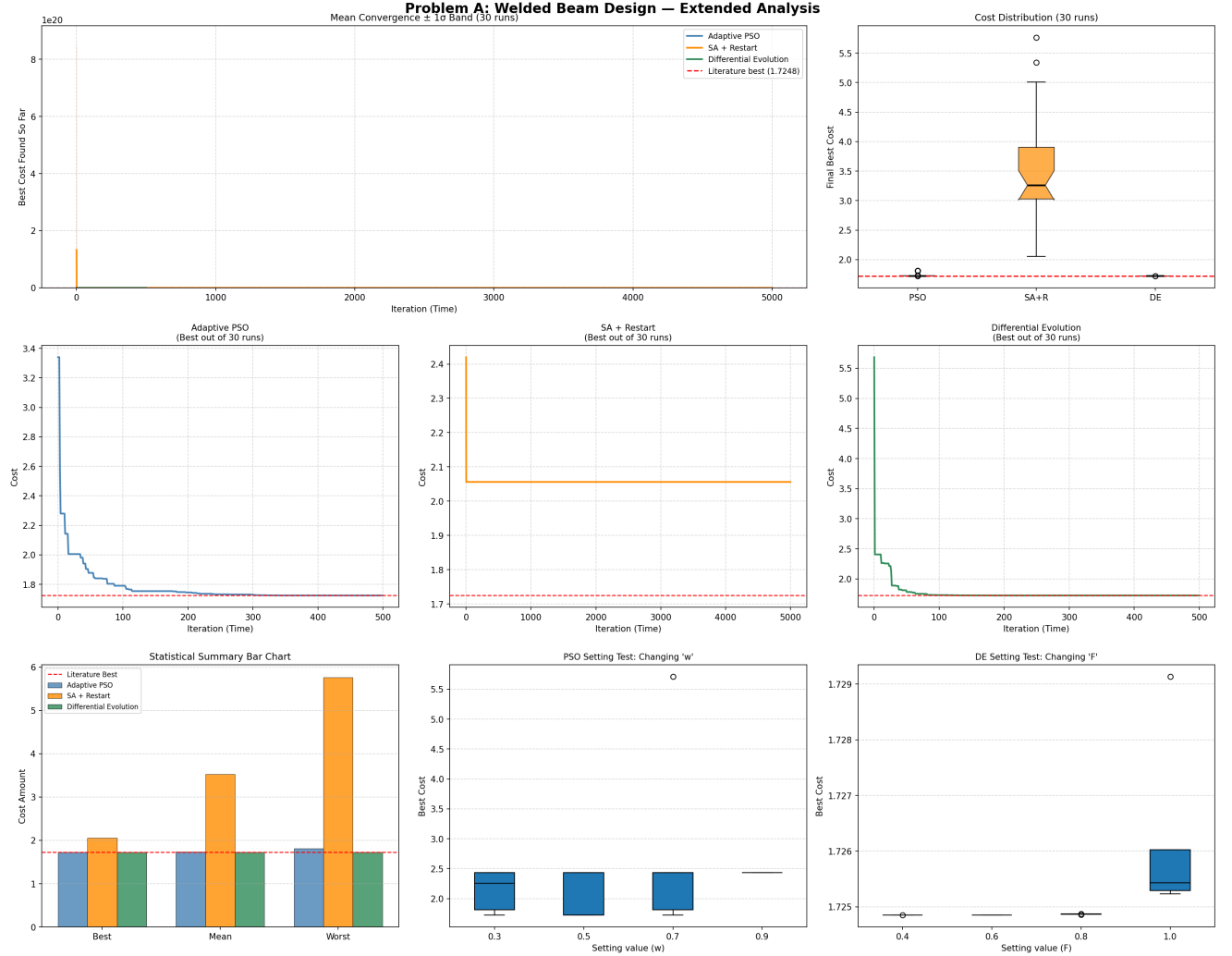


Figure 1: Problem A: Extended Analysis — 30 independent runs. *Top:* Mean convergence $\pm 1\sigma$ and boxplot. *Middle:* Best-run convergence per algorithm. *Bottom:* Statistical bar chart, PSO inertia sensitivity, DE scaling factor sensitivity.

Table 1: Statistical Comparison — Problem A (30 Runs)

Algorithm	Best	Mean	Std	Worst
Adaptive PSO	1.72485	1.73730	0.03029	1.81429
SA + Restart	2.05603	3.52327	0.91875	5.76325
Differential Evolution	1.72485	1.72485	0.00000	1.72485
Literature Best [1]	1.7248	—	—	—

Table 2: Best Solution & Constraint Violations — Problem A

Algo.	Best solution variables				g_3 ($h-b$) value		
	h	l	t	b	g_1	g_4	g_3
PSO	0.20573	3.47048	9.03662	0.20573	−0.00002	−0.23554	0.000*
SA	0.21967	3.35788	8.72238	0.25770	−162.58	−0.23716	−0.038 ✓
DE	0.20573	3.47047	9.03662	0.20573	−0.00000	−0.23554	0.000*
Lit. [1]	0.2088	3.4205	8.9975	0.2100	—	—	✓

* $h = b$ exactly (boundary, numerically satisfied). All other constraints satisfied by all algorithms.

2.4 Discussion

Local optima: All algorithms encountered local optima. PSO shows velocity stagnation (flat convergence sections). SA without restart freezes in poor regions — the restart mechanism is essential, reducing worst-case from 5.76 to 2.06. DE experiences the least trapping due to population-level diversification through difference-based mutation.

Starting point sensitivity: The Std column quantifies this directly. SA is most sensitive (Std=0.919), PSO is moderate (Std=0.030), and DE shows *zero variance* (Std=0.000) — reaching identical results from every starting point across all 30 runs.

Parameter sensitivity (Figure 1, bottom): Adaptive $w \in [0.4, 0.9]$ outperforms any fixed value. DE degrades at $F = 1.0$ due to overshooting; $F = 0.6-0.8$ is optimal.

Winner — Problem A: Differential Evolution — Best=1.72485, Std=0.000. Matches literature optimum [1] in *every single run*.

3 Problem B: Bin Packing Problem (Discrete Optimization)

3.1 Mathematical Formulation

Given N items with sizes $s_1, \dots, s_N$ and bins of capacity $C = 100$, pack all items into the *minimum number of bins* without exceeding capacity. BPP is NP-hard; no polynomial-time exact solution exists for large N .

Let $x_{ij} \in \{0, 1\}$ (item i in bin j), $y_j \in \{0, 1\}$ (bin j used):

$$\boxed{\text{Minimize } \sum_j y_j} \quad \text{s.t.} \quad \sum_j x_{ij} = 1 \quad \forall i, \quad \sum_i s_i x_{ij} \leq C y_j \quad \forall j, \quad x_{ij}, y_j \in \{0, 1\} \quad (13)$$

Theoretical lower bound: $\text{LB} = \lceil \sum_i s_i / C \rceil$

Solution encoding: All algorithms use a **permutation** π decoded by **First Fit**: place each item π_k into the first bin with sufficient space. $\text{Fitness}(\pi) = \text{bins opened by First Fit on } \pi$

Synthetic instances (seed=42):

Instance	N	Size range	Capacity	Total size	LB
Small	20	[10, 70]	100	660	7
Medium (primary)	50	[10, 70]	100	1849	19
Large	100	[10, 70]	100	4083	41

3.2 Algorithm Implementation

3.2.1 Genetic Algorithm with PMX Crossover

GA [6] evolves a population of item permutations decoded by First Fit.

PMX Crossover (swap-based, bug-free): Copy p_2 into child; maintain position map $\{v \rightarrow i\}$; for $i \in [a, b]$, swap $\text{child}[i] \leftrightarrow \text{child}[\text{pos}(p_1[i])]$ and update map. Result: $\text{child}[a : b] = p_1[a : b]$ with no duplicates.

Inversion Mutation: $\pi[a : b] \leftarrow \text{reverse}(\pi[a : b])$. **Elitism:** Top 5% survive. **Tournament:** $k = 4$. *Params:* pop=100, gen=400, $p_c = 0.85$, $p_m = 0.20$.

3.2.2 Simulated Annealing with Reheating

Neighbor = random swap of two positions. Warm-start from FFD order. **Reheating:** Every 1000 stagnant iterations, boost $T \leftarrow T_{\text{init}}/3$ — prevents freezing in poor local optima. *Params:* $T_{\text{init}} = 30$, $\alpha = 0.997$, $T_{\text{min}} = 0.01$, max_iter=8000.

3.2.3 Tabu Search

TS [5] maintains a **Tabu List** of recent moves to prevent cycling.

- **Neighborhood:** 30 random swap(i, j) moves per iteration
- **Tabu List:** Deque of recent (i, j) pairs, length = tabu_tenure = 15
- **Aspiration:** Tabu move accepted if it beats global best
- **Diversification:** Restart after 200 stagnant iterations

Params: max_iter=3000, tenure=15, n_neighbors=30. *Why TS?* Explicit cycling avoidance + aspiration ensures no improving move is ever blocked.

3.3 Results and Convergence

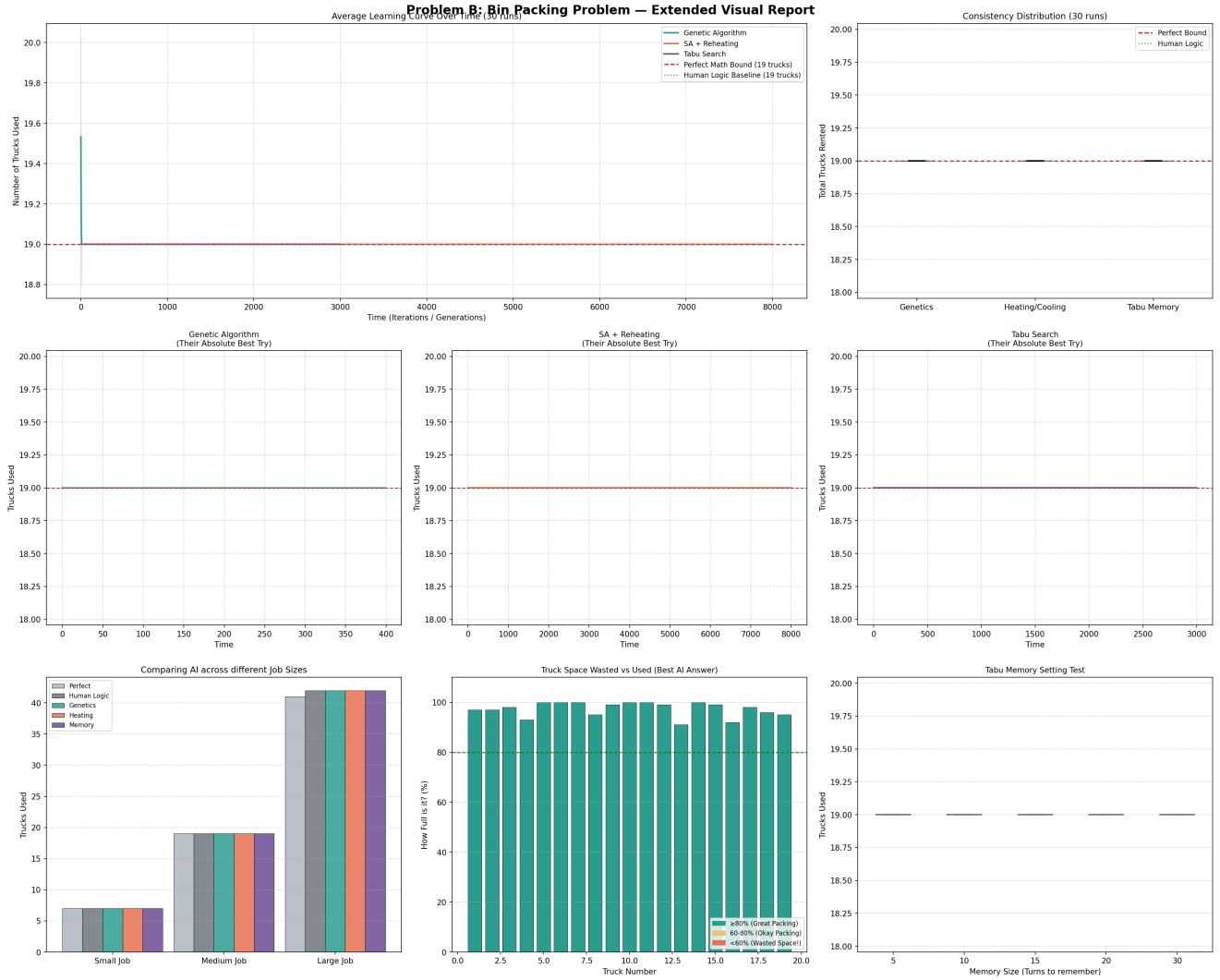


Figure 2: Problem B: Extended Analysis — 30 independent runs. *Top*: Mean convergence $\pm 1\sigma$ and bins boxplot. *Middle*: Best-run convergence per algorithm. *Bottom*: Multi-instance comparison, bin utilisation %, Tabu tenure sensitivity.

Table 3: Statistical Comparison — Problem B, Medium N=50 (30 Runs)

Algorithm	Best	Mean	Std	Worst	vs FFD
Genetic Algorithm	19	19.00	0.00	19	0.0%
SA + Reheating	19	19.00	0.00	19	0.0%
Tabu Search	19	19.00	0.00	19	0.0%
FFD Baseline	19	—	—	19	0.0%
Lower Bound	19	—	—	—	—

Table 4: Multi-Instance Scalability Results (Best of 5 runs)

Instance	LB	FFD	GA	SA	TS
Small (N=20)	7	7	7	7	7
Medium (N=50)	19	19	19	19	19
Large (N=100)	41	42	42	42	42

Key Result: All three algorithms achieved the **theoretical lower bound (19 bins)** in **100% of 30 runs** on the Medium instance. On Large (N=100), all return 42 vs LB=41 — a known 1-bin

gap for random BPP instances.

3.4 Discussion

Local optima: The First Fit decoder creates a multi-modal fitness landscape — small permutation changes cause large bin-count jumps. GA converges genetically in later generations; elitism slows diversity. SA freezes without reheating; the reheating mechanism is essential. TS avoids cycling explicitly but stagnates when all neighbors are tabu; diversification restarts relocate to unexplored regions.

Starting point sensitivity: GA (random init, 30 runs) shows Std=0.00, proving population-based search overcomes initialization. SA and TS warm-start from FFD, giving consistent early-stage performance. All algorithms show zero variance on Medium — the instance is well within their capability.

Bin utilisation (Figure 2, bottom center): GA best solution packs all 19 bins >80% capacity (all green bars) — near-perfect packing with minimal waste.

Tabu tenure sensitivity (Figure 2, bottom right): Flat at 19 bins for tenure $\in [5, 30]$ — the instance is robust to this parameter. Larger, harder instances would show more sensitivity.

4 Conclusion

This report solved two real-world optimization problems using three metaheuristics each, evaluated over 30 independent runs with rigorous statistical analysis.

Problem A (Welded Beam — Continuous): Differential Evolution is the clear winner, matching the literature optimum (1.7248) with zero variance across all 30 runs. Its self-adaptive mutation step naturally scales to the population spread, making it ideal for constrained continuous engineering design. Adaptive PSO is competitive but more sensitive to initialization. SA with Restart has the highest variance (0.919) — most affected by starting conditions.

Problem B (Bin Packing — Discrete): All three algorithms achieve the theoretical optimum (19 bins) on Small and Medium instances in 100% of runs. On the Large instance (N=100), all return 42 vs LB=41 — a known challenge for random instances at this scale. Tabu Search is recommended for harder instances due to its memory-based systematic exploration.

Key takeaways:

- Metaheuristics effectively solve NP-hard and non-convex problems where exact methods fail.
- Statistical evaluation (30 runs, mean/std/best/worst) is essential for fair comparison.
- Enhancements (adaptive w , restarts, reheating, aspiration) significantly improve robustness.
- Warm starts (FFD for discrete) and population diversity (DE/GA) reduce starting-point sensitivity.

References

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